

Notes: Kacperczyk, Pérignon, & Vuillemey (JF, 2021)

The Private Production of Safe Assets

Contents

1	Big picture	3
2	Data and measurement: CDs, T-bills, and safe-asset premia	3
2.1	Why CDs are a good laboratory	3
2.2	CD issuer data	4
2.3	Public safe asset data	4
2.4	Other private short-term assets	4
2.5	Safety premium	4
3	Models and empirical specifications	5
3.1	Theoretical mechanism	5
3.2	Baseline supply response regression	5
3.3	Maturity-choice regression	5
3.4	Stress and fragility	5
3.5	Issuer balance sheets and country characteristics	6
3.6	Instrumental variable strategy	6
4	Identification and empirical design	6
4.1	What identifies the main effect	6
4.2	Why the maturity boundary matters	7
4.3	Why stress periods matter	7
4.4	Addressing alternative explanations	7
4.5	What the paper cannot fully prove	7
5	Tables and figures	7
5.1	Table I, Figure 1, Figure 2, and Table II: market structure and summary statistics	8
5.2	Table III and Figures 3–4: which assets are safe?	8
5.3	Tables IV–VI: public safe-asset supply and private CD issuance	10
5.4	Table VII and Table VIII: market stress and issuer balance sheets	11
5.5	Figure 5, Table IX, and Table X: issuer and country heterogeneity	12
5.6	Table XI, Table XII, and Figure 6: other assets and alternative identification	13

6	Short summary	15
7	Conclusion	16
7.1	Core conclusion	16
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** Can private financial institutions produce assets with safety attributes, and where are the empirical boundaries of that safety?
- **Safe asset definition.** A safe asset is information-insensitive: investors can value it without costly analysis and can use it as a money-like store of value or collateral.
- **Main setting.** The paper studies high-frequency European short-term debt markets from 2008 to 2014, especially euro-denominated certificates of deposit (CDs) issued by banks in the French market.
- **Main data advantage.** The data are security-level and daily, with 1.36 million CD issues by 271 European banks, plus T-bill data, issuer balance-sheet data, and data on other short-term instruments.
- **Main price result.** Only the shortest bank-issued CDs, with maturity less than or equal to one week, carry a safety premium. Their interest rates are about 8 bps per year below the risk-free benchmark.
- **Boundary of private safety.** Longer-term CDs do not look safe: one-month, three-month, and six-month CDs trade above the risk-free benchmark.
- **Main quantity result.** When the supply of public safe assets falls, banks increase issuance of the shortest CDs. In the baseline specification, a 1% decrease in French and German T-bill supply is associated with roughly a 4% increase in very short-term CD issuance.
- **Stress fragility.** During periods of high market stress, the substitution between T-bills and CDs disappears. T-bills keep or strengthen their safety premium, but CDs cease to be perceived as close substitutes.
- **Issuer heterogeneity.** Balance-sheet strength matters less than one might expect. During stress, all CD issuers lose private safety status, not only weak issuers.
- **Country heterogeneity.** The fiscal capacity of the issuer's domicile country is not a first-order determinant of safety perception, although there is evidence of local segmentation in domestic T-bill/CD markets.
- **Asset-type heterogeneity.** Bank-issued CDs are special. Other private short-term assets, especially nonfinancial commercial paper and repos, do not show the same clean private-safe-asset substitution pattern.
- **Economic takeaway.** Private safety is real but narrow and fragile. It is concentrated in very short bank debt and breaks down when investors become worried about information sensitivity.

2 Data and measurement: CDs, T-bills, and safe-asset premia

2.1 Why CDs are a good laboratory

- CDs are unsecured short-term debt securities with maturities from one day to one year.
- The French market covers more than 80% of the global euro-denominated CD market.
- CDs are not deposit-insured, but issuers are generally large banks with strong balance sheets.
- More than 90% of CDs are purchased by money market funds, which are natural safety-demand investors.
- The same issuer can issue CDs at many maturities on the same day, allowing within-issuer comparisons across maturities.

Interpretation. The core empirical advantage is granularity. The paper can compare short and long CDs issued by the same bank in the same week, which helps isolate safety-demand supply responses.

2.2 CD issuer data

- The Banque de France provides daily issuance data for euro-denominated CDs issued by European banks.
- The sample covers 271 issuing banks and 1,360,272 CD issues from 2008 to 2014.
- France accounts for 72.3% of issuers and 72.8% of issuance by volume.
- CDs are economically meaningful for banks: conditional on positive CD balances, CDs average 9% of total liabilities and 16% of short-term debt.

Interpretation. The market is not a small niche. It is a major bank funding market and therefore an appropriate setting for studying private safe-asset production.

2.3 Public safe asset data

- The public safe asset benchmark is euro-denominated T-bills, primarily from France and Germany.
- France and Germany are the largest core euro-area T-bill issuers and issue at weekly frequency.
- Italy and Spain are used as a counterfactual public asset supply measure because their T-bills do not exhibit the same safety-premium pattern.

Interpretation. The identifying logic is that investors treat French and German T-bills as safe public assets. If private issuers cater to excess safe-asset demand, their response should be strongest when these T-bills become scarcer.

2.4 Other private short-term assets

- The paper also collects euro-denominated commercial paper issued by nonfinancial firms.
- For France and Germany, the paper uses national financial accounts to examine repos, short-term debt of other financial institutions, and nonfinancial commercial paper.

Interpretation. This lets the authors test whether private safety is a general property of short-term debt or specific to bank-issued CDs.

2.5 Safety premium

The safety premium is the spread between an asset's interest rate and a risk-free benchmark that does not itself provide money-like convenience:

$$SP_{TB} = r_{TB} - r_f, \quad SP_{CD} = r_{CD} - r_f. \quad (1)$$

A security has a safety premium when this value is negative.

- r_f is measured using the Eonia overnight indexed swap rate at the same maturity.
- Eonia swap rates are low-credit-risk, liquid, and not directly pledgeable collateral, so they are used as a risk-free benchmark without safe-asset convenience yield.
- The paper matches CD and T-bill rates to swap rates of the same maturity.

Interpretation. Negative safety premia mean issuers can borrow below the risk-free benchmark because investors are willing to pay for safety, liquidity, and information insensitivity.

3 Models and empirical specifications

3.1 Theoretical mechanism

The core mechanism is:

Public safe-asset shortage $\Rightarrow$ excess safety demand $\Rightarrow$ private issuance of the safest private substitutes.

- **Hypothesis 1.** Private assets can benefit from a safety premium.
- **Hypothesis 2.** Higher demand for safety is associated with higher issuance of private safe assets.

Interpretation. Hypothesis 1 is a price test. Hypothesis 2 is a quantity test. The paper’s central contribution is that it tests both and then maps where the relationship fails.

3.2 Baseline supply response regression

The baseline quantity specification is

$$\Delta \log(1 + Q_{CD,i,mat,t}) = \phi \mu_{i,mat} \Delta \log(Q_{TB,t}) + \rho' Controls_{i,t-1} + \mu_{i,t} + \mu_{i,mat} + \varepsilon_{i,mat,t}. \quad (2)$$

- $Q_{CD,i,mat,t}$ is CD issuance by issuer i , maturity bucket mat , and week t .
- $Q_{TB,t}$ is T-bill issuance by France and Germany.
- Maturity buckets are below one week, one week to one month, one month to three months, and above three months.
- Issuer-date fixed effects absorb total issuer-week demand and issuer-level shocks.
- Issuer-maturity fixed effects absorb persistent maturity preferences of each issuer.

Interpretation. The key coefficient is the maturity-specific response to T-bill supply. A negative coefficient for very short CDs means that when safe public asset supply falls, banks issue more short CDs.

3.3 Maturity-choice regression

The paper also studies whether issuers change the maturity composition of CDs:

$$ShortIssuance_{i,t}^{CD} = \beta \Delta \log(Q_{TB,t}) + \phi \Delta \log(Q_{TB,t}) HighStress_t + \rho' Controls_{i,t-1} + \mu_i + \mu_q + \varepsilon_{i,t}. \quad (3)$$

Interpretation. This specification asks whether banks shorten maturities when safety demand changes, rather than only changing the total amount of short CDs.

3.4 Stress and fragility

High stress is defined using either:

- VIX above the 75th percentile, or
- Euribor–Eonia spread above the 75th percentile.

The baseline CD issuance response is interacted with $HighStress_t$.

Interpretation. If private safety is fragile, the short-CD response should vanish in stress periods. This is exactly what the paper finds.

3.5 Issuer balance sheets and country characteristics

The paper tests whether the substitution effect varies with issuer characteristics:

- size;
- equity/assets;
- impaired loans/loans;
- return on assets;
- G-SIB status.

It also tests country fiscal capacity using a low-capacity dummy based on average sovereign CDS spreads above 200 bps.

Interpretation. These tests ask whether investors evaluate safety at the issuer level, country level, or asset-class level. The stress results point strongly toward asset-class-level fragility.

3.6 Instrumental variable strategy

The IV test uses the T-bill bid-to-cover ratio:

$$BTC_t = \frac{\text{bids for T-bills at } t}{\text{T-bills allotted at } t}. \quad (4)$$

The first stage is

$$SP_{CD,t} = \eta_1 + \pi_1 BTC_t + \varepsilon_{1,t}, \quad (5)$$

and the second stage is

$$Q_{CD} = \eta_2 + \pi_2 \widehat{SP}_{CD} + \varepsilon_2. \quad (6)$$

Interpretation. High bid-to-cover ratios measure excess public safe-asset demand. If this demand lowers private safe-asset yields and increases short-CD issuance, the IV strategy supports a demand-driven interpretation.

4 Identification and empirical design

4.1 What identifies the main effect

- The baseline uses within-issuer, within-week variation across maturities.
- Issuer-date fixed effects absorb each bank's total funding need in that week.
- Issuer-maturity fixed effects absorb each bank's persistent preference for particular maturities.
- The remaining variation asks whether issuance shifts specifically toward the maturities investors perceive as safe when public safe assets are scarce.

Interpretation. The design is powerful because it avoids comparing different banks in different weeks. It compares the same bank's issuance mix across maturity buckets.

4.2 Why the maturity boundary matters

- The price evidence says safety is concentrated in one-week CDs.
- The quantity evidence says the response to T-bill scarcity is also concentrated in very short CDs.
- Longer CDs do not carry a safety premium and do not respond similarly to public safe-asset scarcity.

Interpretation. The fact that price and quantity boundaries line up is a central validation of the safe-asset interpretation.

4.3 Why stress periods matter

- In stress periods, investors become more concerned about adverse selection.
- T-bills remain safe because they are backed by governments.
- Private short-term debt becomes information-sensitive.
- The CD market does not collapse, but CDs cease to substitute for T-bills.

Interpretation. This distinguishes a market-volume story from a safety-perception story. CDs are still issued, but investors no longer treat them as safe substitutes.

4.4 Addressing alternative explanations

- **Crowding out.** A generic crowding-out story should affect many maturities, not only the shortest CDs.
- **Gap filling.** A maturity clientele story should also operate across longer maturities, but the effect is concentrated in short CDs.
- **Issuer selection.** Issuer-date fixed effects absorb time-varying issuer-level funding demand.
- **Public asset quality.** Italy and Spain T-bills do not generate the same substitution pattern, supporting the safe-public-asset interpretation.
- **Causal demand pressure.** The bid-to-cover IV and auction-day evidence link the mechanism to excess public safe-asset demand.

4.5 What the paper cannot fully prove

- It does not observe investor identity in every CD transaction.
- It cannot fully separate safety from liquidity, since the two are closely connected in theory.
- It does not explain why aggregate CD volumes did not collapse during stress.
- It studies euro-denominated European short-term markets, so external validity to other private safe assets must be assessed separately.

5 Tables and figures

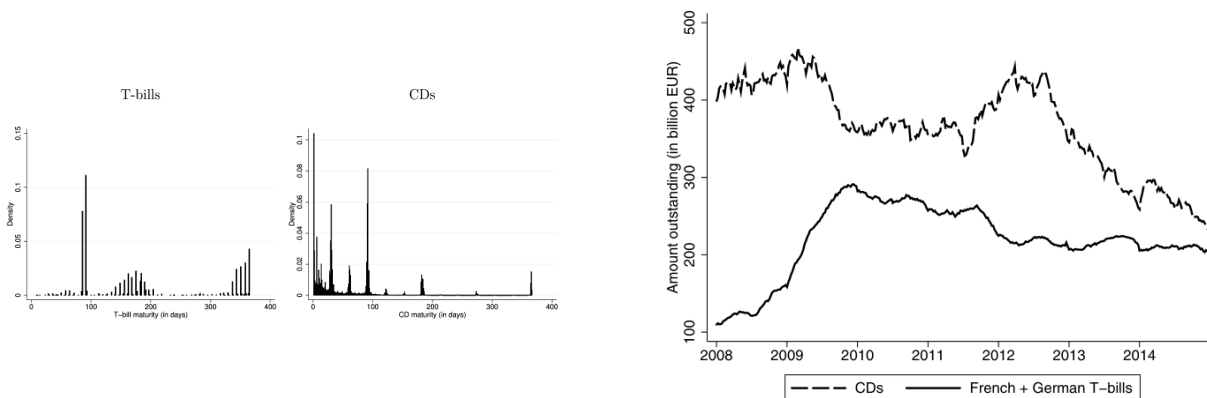
This section uses the same *Read plus Economic message* format. Adjacent visuals are grouped to save space and improve comparison.

5.1 Table I, Figure 1, Figure 2, and Table II: market structure and summary statistics

Read:

- Table I shows that the CD market is dominated by bank issuers, especially French banks, but includes banks from many European countries.
- The average CD issuer has substantial regulatory capital: mean Tier-1 capital is 13.07%, and mean total regulatory capital is 16.12%.
- CDs are meaningful funding instruments: conditional on CD use, CDs average 9% of total liabilities and 16% of short-term debt.
- Figure 1 shows maturity clustering for CDs at one day, one week, one month, and three months; T-bills cluster around three months.
- Figure 2 shows that the CD market is larger than the French and German T-bill market over the sample, though CD volumes decline after 2012.
- Table II shows average amounts outstanding of EUR 366.4 billion for CDs and EUR 220.1 billion for T-bills.

Economic message. The market is large, high frequency, and maturity-rich. This makes it possible to identify private safety production at a granular level.



5.2 Table III and Figures 3–4: which assets are safe?

Read:

- Table III shows that one-week CDs have a safety premium of -8.1 bps, meaning banks borrow below the Eonia risk-free benchmark.
- One-month T-bills have a larger safety premium of -16.2 bps.
- CDs above one week have positive spreads over the risk-free benchmark: 11.5 bps at one month, 23.9 bps at three months, and 37.4 bps at six months.
- The CD safety-premium term structure steepens most sharply between one week and one month, exactly where private safety disappears.
- Figure 3 shows that CD safety premia tend to vanish during stress, while T-bill premia become stronger.
- Figure 4 shows that the difference between CD and T-bill premia widens during stress.

Table I
Description of the Sample of CD Issuers

This table describes the sample of CD issuers. Panel A shows the shares of issuers and of issued CD amounts by country. It also reports the average five-year credit default swap (CDS) spread of each country over our sample period (2008 to 2014). Panel B provides descriptive statistics on the distribution of CD issuers' balance sheet characteristics. Means and quantiles are as of the end of December of each year and are computed from the pooled sample over the period 2008 to 2014. Panel C relates CD amounts outstanding as of the end of December of each year to other balance sheet characteristics, in the pooled sample. Statistics are conditional on the issuer having nonzero CDs outstanding. The calculation of CD/(CD + Repo) is also conditional on the issuer having nonzero repurchase agreements outstanding. All variables are defined in Table IAI of the Internet Appendix.

Panel A: Geographic Distribution of Issuers								
	Banks		% Issued		Avg. Country			
	# Issuers	% Issuers	Amount	Largest Issuer	CDS (bps)			
All	271	100.00	100.00	—	—			
Austria	2	0.74	0.20	Oesterreich. Kontrollbank	77			
Belgium	2	0.74	6.26	Dexia Credit Local	105			
Denmark	3	1.11	0.56	Jyske Bank	52			
France	196	72.32	72.83	BNP Paribas	76			
Germany	12	4.43	1.08	HypoVereinsbank	41			
Ireland	7	2.58	0.48	Allied Irish Banks	318			
Italy	14	5.17	3.18	Unicredit	202			
Netherlands	8	2.95	5.42	Rabobank	54			
Spain	2	0.74	0.58	BBVA	203			
Sweden	4	1.48	0.89	Svenska Handelsbanken	38			
Switzerland	2	0.74	0.44	UBS	55			
UK	11	4.06	7.36	HSBC	58			
Others	8	2.95	1.12	—	—			

Panel B: Balance Sheet Characteristics								
	10 th	25 th	Mean	Median	75 th	90 th	Std.	Obs.
Size (log Total assets)	20.82	22.07	23.50	23.34	24.71	26.70	2.09	1,449
Loans/Assets	0.27	0.48	0.63	0.69	0.82	0.88	0.23	1,445
Customer deposits/Assets	0.03	0.20	0.37	0.35	0.57	0.66	0.23	1,422
Short-term debt/Assets	0.05	0.13	0.29	0.24	0.48	0.60	0.21	1,422
ROA (%)	-0.20	0.15	0.32	0.40	0.74	1.04	1.15	1,443
ROE (%)	-3.88	2.52	3.57	5.42	8.32	13.27	12.36	1,443
Net interest margin Assets	0.00	0.01	0.01	0.01	0.02	0.03	0.01	1,411
Impaired loans/Loans (%)	1.04	2.24	5.42	3.91	6.59	11.89	5.08	1,056
Equity/Assets	0.03	0.04	0.08	0.07	0.11	0.13	0.05	1,449
Tier-1 capital (%)	7.60	9.20	13.07	11.20	14.30	18.25	7.29	458
Total regulatory capital (%)	9.90	11.60	16.12	13.70	16.91	21.4	10.27	486

Panel C: Size of CD Funding								
CD/Equity	0.01	0.05	1.17	0.21	0.69	2.25	0.33	971
CD/(CD + Repo)	0.01	0.05	0.34	0.22	0.61	0.85	0.39	218
CD/Short-term debt	0.00	0.01	0.16	0.05	0.18	0.49	0.23	971
CD/Total liabilities	0.00	0.01	0.09	0.03	0.09	0.22	0.10	1,007

Table II
Descriptive Statistics on Short-Term Debt Securities

This table describes our data on the issuance of short-term debt securities over our sample period (2008 to 2014). In Panels A, B, and C, the universe of assets includes T-bills issued by the French and German Treasuries and CDs issued by European banks. Panel A shows the amount of securities outstanding. Panel B shows net issuances, defined as the change in amounts outstanding between Fridays of two consecutive weeks. Panel C shows the maturity of new issues, measured in days, both unweighted and weighted by the issue amount. Unweighted moments are computed based on the sample of all issuances. Weighted moments are computed as averages by day. Panel D shows the amounts of other securities outstanding.

Panel A: Aggregate T-Bill and CD Amounts Outstanding (in EUR billion)											
	Min	10pc	25pc	Median	Mean	75pc	90pc	Max	Std.	Obs.	
Total T-bill outstanding	109.6	139.2	207.4	218.9	220.1	257.1	273.2	291.2	45.4	364	
Total CD outstanding	241.0	275.5	333.7	369.1	366.4	419.9	433.4	467.6	58.2	364	
CD outstanding ≤ 1w	1.1	8.9	18.0	28.4	27.4	35.4	44.6	64.4	13.4	364	
CD outstanding ∈ (1w, 1m]	2.7	8.3	12.0	18.0	19.0	23.4	32.1	50.0	9.2	364	
CD outstanding ∈ (1m, 3m]	25.7	49.7	67.4	142.1	134.8	189.9	225.0	243.7	63.4	364	
CD outstanding > 3m	108.5	134.3	161.2	176.0	185.1	210.9	241.7	276.3	39.4	364	

Panel B: Aggregate T-Bill and CD Net Issuance (in EUR billion)											
	Min	10pc	25pc	50pc	Mean	75pc	90pc	Max	Std.	Obs.	
Total T-bill net issuance	-7.5	-2.3	-0.9	0.2	0.2	1.5	2.5	7.2	2.0	363	
Total CD net issuance	-21.8	-7.2	-3.8	-0.5	-0.5	2.8	6.3	23.8	5.8	363	
CD net issuance ≤ 1w	-11.4	-4.7	-2.4	-0.1	-0.1	1.7	4.6	12.7	3.7	363	
CD net issuance ∈ (1w, 1m]	-8.1	-2.9	-1.3	0.1	-0.1	1.0	2.2	8.6	2.2	363	
CD net issuance ∈ (1m, 3m]	-15.1	-4.8	-2.9	-0.3	-0.5	1.8	4.1	16.7	4.0	363	
CD net issuance > 3m	-9.8	-3.7	-1.3	0.2	0.2	2.1	3.8	17.3	3.1	363	

Panel C: Maturity of New Issues (in Days)											
	Min	10pc	25pc	50pc	Mean	75pc	90pc	Max	Std.	Obs.	
<i>Pooled Data</i>											
T-bill	7	84	91	154	185	337	357	365	111	1,145	
CD	1	2	13	33	66	92	181	367	76	841,636	
<i>Volume Weighted (Daily)</i>											
T-bill	53	129	154	164	161	173	182	227	23	359	
CD	5	13	18	26	40	39	79	365	44	2,185	

Panel D: Aggregate Amounts Outstanding of Other Securities (in EUR billion)											
	Min	10pc	25pc	Median	Mean	75pc	90pc	Max	Std.	Obs.	
<i>France</i>											
Other financials	0	0	5.5	25.2	22.9	34.9	58.1	77.0	21.4	96	
Commercial paper	16.3	22.1	27.3	32.4	34.8	39.5	50.8	64.7	10.7	96	
Repos	215.1	266.6	297.9	614.1	570.3	713.8	768.1	1630.2	299.4	96	
<i>Germany</i>											
Other financials	0	0	0	2.5	2.8	4.4	6.2	14.7	3.1	84	
Commercial paper	2.7	4.4	7.9	15.2	16.4	22.7	28.7	46.0	10.2	84	
Repos (monthly)	9.1	23.7	52.5	79.4	89.8	132.2	160.9	188.5	49.1	252	

Economic message. The price evidence draws a sharp boundary: private safety exists for the shortest bank CDs, not for longer CDs. During stress, investors flee from private short-term debt toward public safe assets.

Table III
Safety Premium on T-Bills and CDs

This table displays the safety premium on T-bills and CDs with maturities between one week and one year. The safety premium is defined by equation (1). The T-bill safety premium is computed as the weighted average (by outstanding volume) of the French and the German safety premia. In Panel A, the safety premium is computed over the full sample period (2008 to 2014) for each maturity. In each subpanel, the last column shows the difference between the safety premium at the longest and the shortest available maturities. In Panel B, we display linear slopes of the weekly CD safety premium over three intervals: [1 week; 1 month], [1 month; 3 months], and [3 months; 6 months]. The first three columns display the average slopes and their standard deviations. The last two columns report differences and standard deviations obtained from a two-sample *t*-test. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Full Sample Period										
T-Bills				CD						
1m	3m	6m	12m	12m-1m	1w	1m	3m	6m	6m-1w	
-0.162***	-0.128***	-0.072***	-0.025***	0.137***	-0.081***	0.115***	0.239***	0.374***	0.461***	
(0.008)	(0.008)	(0.006)	(0.006)	(0.008)	(0.005)	(0.036)	(0.024)	(0.025)	(0.025)	
Obs. 338	338	338	338	338	338	338	338	338	338	

Panel B: Slope of CD Safety Premium				
Average Slopes		Differences in Slopes		
[1w, 1m]	[1m, 3m]	[3m, 6m]	[1w, 1m] - [1m, 3m]	[1m, 3m] - [3m, 6m]
0.049***	0.024***	0.017***	0.024***	0.008***
(0.003)	(0.001)	(0.001)	(0.002)	(0.001)

Table 1: Safety premia on T-bills and CDs

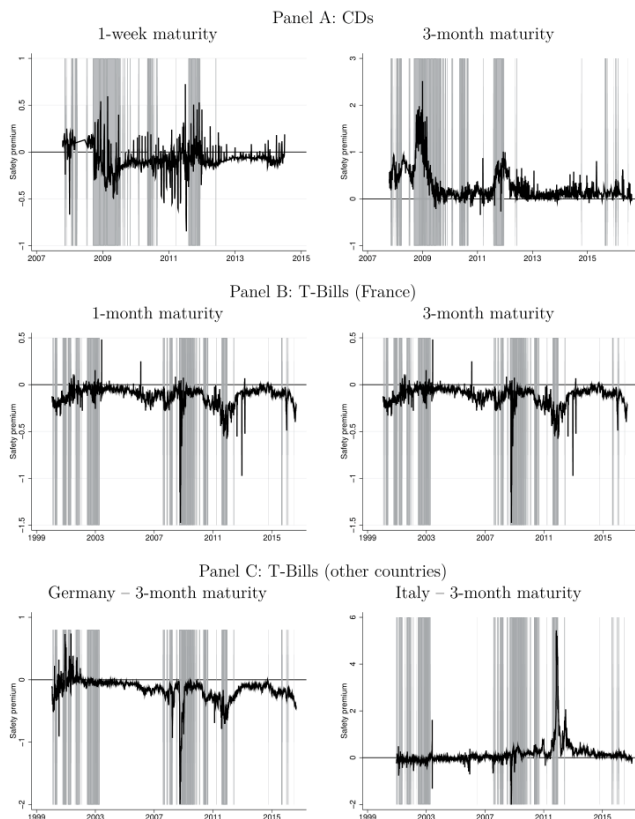


Figure 1: Safety premium across maturities and countries

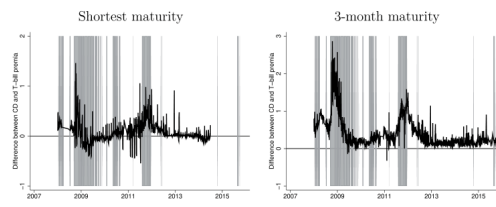


Figure 4. Difference between CD and T-bill safety premia. This figure plots the difference

Figure 2: Difference between CD and T-bill safety premia

5.3 Tables IV–VI: public safe-asset supply and private CD issuance

Read:

- Table IV Panel A shows a strong negative relation between French/German T-bill issuance and issuance of CDs with maturity below one week.
- With issuer-date and issuer-maturity fixed effects, the coefficient for very short CDs is about -4.035 , implying that a 1% fall in public T-bill supply is associated with about a 4% rise in very short CD issuance.
- The coefficient for one-week to one-month CDs is smaller and less robust; longer maturities are insignificant.
- Table IV Panel B uses Italian and Spanish T-bills as a counterfactual public asset measure and finds no significant relation.
- Table V shows that T-bill scarcity affects safety premia: lagged T-bill issuance is positively related to changes in short T-bill and short CD premia, consistent with safety premia becoming more negative when public safe assets are scarce.
- Table VI shows that issuance composition changes mainly at the very short maturity boundary.

Economic message. The quantity response occurs where the price response says safety exists. Banks produce more very short CDs when safe public assets become scarce.

Table IV
T-Bill Issuance and CD Issuance: Baseline Estimation

This table presents results from our baseline estimation of the effect of T-bill issuance on CD issuance. In Panel A, the measure of public asset issuance is the sum of T-bill issuances by France and Germany, while in Panel B it is T-bill issuances by Italy and Spain. The measure of private asset issuance includes all CD issuers in Europe. Regressions include issuer*date fixed effects and thus isolate variation across maturities for a given issuer. Standard errors are clustered at the issuer and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Baseline Estimation

	Dependent Variable: $\Delta \log(1 + Q_{CD,i,mat,t})$						
$\Delta Q_{TB,t} \cdot \mathbb{1}(mat \leq 1w)$	-2.946** (1.205)	-3.941*** (1.255)	-4.035*** (1.270)	-5.603*** (2.138)	-6.926*** (2.048)	-7.140*** (2.112)	-4.184*** (1.422)
$\Delta Q_{TB,t} \cdot \mathbb{1}(1w < mat \leq 1m)$	-1.348 (0.883)	-2.343** (0.900)	-2.234** (0.927)	-0.452 (1.605)	-1.775 (1.706)	-1.977 (1.767)	-1.601 (1.001)
$\Delta Q_{TB,t} \cdot \mathbb{1}(1m < mat \leq 3m)$	0.787 (0.664)	-0.207 (0.711)	-0.017 (0.721)	2.441** (1.170)	1.118 (1.298)	0.845 (1.318)	0.596 (0.829)
R^2	0.000	0.252	0.252	0.000	0.252	0.252	0.252
Obs.	396,384	396,384	396,384	283,920	283,920	283,920	396,384
Excl. 2008 to 2009	No	No	No	Yes	Yes	Yes	No
French T-bills only	No	No	No	No	No	No	Yes
Issuer* Maturity FE	No	No	Yes	No	No	Yes	Yes
Issuer* Date FE	No	Yes	Yes	No	Yes	Yes	Yes

Panel B: Public Safe Asset Measured with Italy and Spain

	Dependent Variable: $\Delta \log(1 + Q_{CD,i,mat,t})$		
$\Delta Q_{TB,t} \cdot \mathbb{1}(mat \leq 1w)$	-0.728 (1.458)	-0.111 (1.595)	-0.104 (1.158)
$\Delta Q_{TB,t} \cdot \mathbb{1}(1w < mat \leq 1m)$	-0.694 (1.108)	-0.076 (1.140)	-0.102 (1.151)
$\Delta Q_{TB,t} \cdot \mathbb{1}(1m < mat \leq 3m)$	-0.291 (0.725)	0.325 (0.876)	0.268 (0.880)
R^2	0.000	0.252	0.252
Obs.	283,920	283,920	283,920
Issuer* Maturity FE	No	No	Yes
Issuer* Date FE	No	Yes	Yes

Table 2: Baseline T-bill/CD issuance regression

Table V
T-Bill Issuance and Safety Premia

This table explains the dynamics of T-bill and CD safety premia by regressing them on lagged T-bill issuance, for various maturities. The T-bill safety premium is computed as the weighted average (by outstanding volume) of the French and German safety premia. We further include month fixed effects. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable: Change in T-Bill Safety Premium					
	1m	3m	6m	9m	12m
$\Delta Q_{TB,t-1}$	1.412** (0.708)	1.318** (0.611)	0.326 (0.424)	0.035 (0.291)	0.315 (0.353)
R^2	0.106	0.162	0.236	0.260	0.267
Obs.	336	336	336	336	336
Month FE	Yes	Yes	Yes	Yes	Yes

Dependent Variable: Change in CD Safety Premium						
	1w	1m	3m	6m	9m	12m
$\Delta Q_{TB,t-1}$	1.073* (0.682)	0.532 (0.765)	0.473 (0.812)	-0.753 (1.203)	-0.898 (3.080)	-2.020 (3.176)
R^2	0.048	0.214	0.165	0.159	0.293	0.177

Table 3: T-bill issuance and safety premia

Table VI
T-Bill Issuance and CD Maturity: Time-Series Variation

This table presents results from estimation of the effect of T-bill issuance on CD maturity. Regressions are at the issuer level. The measure of public asset issuance is the sum of T-bill issuances by France and Germany. The measure of private asset issuance includes all CD issuers in Europe. CD maturity is measured as the share of CDs issued with a maturity of less than one week (left panel) or with a maturity between one week and one month (right panel). Standard errors are clustered at the issuer and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable: Share of CDs Issued with Maturity						
$\Delta \log(Q_{TB,t})$	Below 1 Week			Between 1 Week and 1 Month		
	0.453* (0.230)	-0.396** (0.165)	-0.390*** (0.143)	0.273 (0.260)	0.071 (0.258)	0.109 (0.245)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.552	0.007	0.559	0.228	0.016	0.231
Observations	43,366	43,372	43,366	43,366	43,372	43,366
Issuer FE	Yes	No	Yes	Yes	No	Yes
Quarter FE	No	Yes	Yes	No	Yes	Yes

Table 4: T-bill issuance and CD maturity

5.4 Table VII and Table VIII: market stress and issuer balance sheets

Read:

- Table VII shows that the negative short-CD response to T-bill scarcity is offset during high-stress periods.
- Using the VIX stress measure, the very short CD coefficient is around -7.63 in the full fixed-effect specification, while the interaction with high stress is about $+7.54$, nearly canceling the response.
- The same qualitative result appears using the Euribor–Eonia spread as the stress measure.
- Table VIII Panel A shows some evidence that stronger issuers respond more in normal periods.

- Table VIII Panel B shows that, once stress arrives, balance-sheet strength does not preserve the safe-asset substitution relation.

Economic message. Private safety is fragile at the asset-class level. Even strong banks do not continue to produce assets perceived as safe substitutes during market stress.

Table VII
T-Bill Issuance and CD Issuance: Time-Series Variation

This table presents results from estimation of the effect of T-bill issuance on CD issuance, after including time-series variation in the effect. Regressions are at the issuer level. The measure of public asset issuance is the sum of T-bill issuances by France and Germany. The measure of private asset issuance includes all CD issuers in Europe. Regressions include issuer-date fixed effects and thus isolate variation across maturities for a given issuer. The baseline coefficients are interacted with a *HighStress* dummy that takes the value of one if either the VIX (columns (1) to (3)) or the Euribor-Eonia spread (columns (4) to (6)) is above the 75th percentile of their distribution. Standard errors are clustered at the issuer and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: $\Delta \log(1 + Q_{CD,mat,t})$					
	Stress Measure: VIX			Stress Measure: Euribor-Eonia Spread		
$\Delta Q_{TBL} \cdot 1(mat \leq 1w)$	-6.187*** (1.689)	-7.625*** (1.669)	-7.630*** (1.674)	-3.621** (1.570)	-5.083*** (1.505)	-5.159*** (1.518)
$\Delta Q_{TBL} \cdot 1(mat \leq 1w) \cdot HighStress_t$	6.593*** (2.513)	7.467*** (2.503)	7.537*** (2.583)	1.693 (2.676)	2.893 (2.837)	2.863 (2.823)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m)$	-1.075 (1.355)	-2.514* (1.461)	-2.533* (1.457)	-1.559 (1.205)	-3.021** (1.273)	-2.922** (1.293)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m) \cdot HighStress_t$	-0.526 (1.707)	0.347 (1.768)	0.628 (1.792)	0.517 (1.905)	1.717 (2.005)	1.754 (2.001)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m)$	2.253** (1.034)	0.814 (1.161)	0.788 (1.145)	1.503 (0.912)	0.042 (0.998)	0.205 (0.994)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m) \cdot HighStress_t$	-2.944** (1.272)	-2.071 (1.425)	-1.691 (1.460)	-1.831 (1.218)	-0.631 (1.314)	-0.569 (1.288)
R ²	0.000	0.252	0.252	0.000	0.252	0.252
Obs.	396,384	396,384	396,384	396,384	396,384	396,384
Issuer* <i>Maturity</i> FE	No	No	Yes	No	No	Yes
Issuer* <i>Date</i> FE	No	Yes	Yes	No	Yes	Yes

Table 5: Time-series variation in the CD issuance response

T-Bill Issuance and CD Issuance: Cross-Sectional Variation

This table presents results from estimation of the effect of T-bill issuance on CD issuance, after including cross-sectional variation in the effect. Regressions are at the issuer level. The measure of public asset issuance is the sum of T-bill issuances by France and Germany. The measure of private asset issuance includes all CD issuers in Europe. Regressions include issuer-date fixed effects and thus isolate variation across maturities for a given issuer. The baseline coefficients are interacted with a *HighBS* dummy that takes the value of one if a given balance-sheet variable is above the sample median. In Panel B, we further interact these coefficients with a *HighVIX* dummy that takes the value of one if the VIX is above the 75th percentile of its distribution. Standard errors are clustered at the issuer and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: $\Delta \log(1 + Q_{CD,mat,t})$				
	Size	Equity /Assets	Balance-Sheet Measure: Impaired /Loans	ROA	G-SIB
$\Delta Q_{TBL} \cdot 1(mat \leq 1w)$	-3.636** (1.400)	-3.839** (1.808)	-4.908* (2.521)	-3.039 (2.219)	-2.589** (1.296)
$\Delta Q_{TBL} \cdot 1(mat \leq 1w) \cdot HighBS_{i,t}$	-1.096 (2.000)	-0.557 (2.536)	0.831 (2.664)	-2.154 (1.703)	-4.071** (1.703)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m)$	-2.866* (1.438)	-1.850 (1.372)	-3.728* (2.187)	0.108 (1.435)	-1.682* (0.890)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m) \cdot HighBS_{i,t}$	1.215 (2.209)	-0.557 (1.797)	2.686 (2.655)	-1.150 (2.297)	-1.550 (1.550)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m)$	-0.113 (1.040)	0.955 (1.115)	2.489* (1.561)	0.981 (1.315)	0.981 (0.841)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m) \cdot HighBS_{i,t}$	1.056 (1.472)	-1.065 (1.344)	2.328 (1.973)	-3.719** (1.531)	-2.812** (1.129)
R ²	0.253	0.253	0.253	0.252	0.252
Obs.	300,552	300,552	219,136	299,304	396,384
Issuer* <i>Maturity</i> FE	Yes	Yes	Yes	Yes	Yes
Issuer* <i>Date</i> FE	Yes	Yes	Yes	Yes	Yes

(Continued)

Table VIII—Continued

Panel B: Balance Sheet Characteristics over Time

	Dependent Variable: $\Delta \log(1 + Q_{CD,mat,t})$				
	Size	Equity /Assets	Balance-Sheet Measure: Impaired /Loans	ROA	G-SIB
$\Delta Q_{TBL} \cdot 1(mat \leq 1w)$	-6.659*** (1.990)	-10.313*** (3.143)	-10.904*** (3.766)	-7.647** (3.840)	-5.366*** (1.844)
$\Delta Q_{TBL} \cdot 1(mat \leq 1w) \cdot HighVIX_t$	6.209*** (2.758)	13.945*** (3.801)	13.915*** (4.973)	9.469* (5.039)	5.821** (2.728)
$\Delta Q_{TBL} \cdot 1(mat \leq 1w) \cdot HighBS_{i,t}$	-3.354 (3.506)	3.779 (3.592)	4.150 (4.178)	-1.483 (4.648)	-6.372** (3.128)
$\Delta Q_{TBL} \cdot 1(mat \leq 1w) \cdot HighVIX_t \cdot HighBS_{i,t}$	5.143 (3.955)	-8.648** (3.718)	-7.932 (5.403)	-1.050 (5.604)	4.828 (4.063)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m)$	-3.727 (2.407)	-3.645 (2.382)	-4.007** (2.910)	-1.696 (2.565)	-1.696 (1.509)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m) \cdot HighVIX_t$	-0.262 (2.841)	3.408 (2.480)	3.296 (4.046)	1.253 (2.956)	0.008 (1.770)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m) \cdot HighBS_{i,t}$	-0.702 (2.253)	1.106 (2.242)	4.105 (5.458)	-4.569 (4.392)	-2.357 (3.002)
$\Delta Q_{TBL} \cdot 1(1w < mat \leq 1m) \cdot HighVIX_t \cdot HighBS_{i,t}$	4.111 (4.349)	-3.403 (3.242)	-3.433 (5.458)	0.789 (4.392)	1.693 (3.002)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m)$	1.601 (1.441)	2.553 (1.846)	-0.770 (2.021)	4.763** (2.162)	1.756 (1.335)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m) \cdot HighVIX_t$	-3.562* (1.869)	-3.270 (2.238)	0.485 (2.994)	-4.658* (2.658)	-1.625 (1.894)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m) \cdot HighBS_{i,t}$	0.063 (1.963)	-1.808 (2.142)	4.263* (2.443)	-5.519** (2.401)	-2.722 (1.765)
$\Delta Q_{TBL} \cdot 1(1m < mat \leq 3m) \cdot HighVIX_t \cdot HighBS_{i,t}$	1.979 (2.922)	1.422 (3.890)	-4.319 (3.123)	3.658 (3.123)	-0.185 (2.344)
R ²	0.253	0.253	0.253	0.252	0.252
Obs.	300,552	300,552	219,136	299,304	396,384
Issuer* <i>Maturity</i> FE	Yes	Yes	Yes	Yes	Yes
Issuer* <i>Date</i> FE	Yes	Yes	Yes	Yes	Yes

5.5 Figure 5, Table IX, and Table X: issuer and country heterogeneity

Read:

- Figure 5 shows that stress and non-stress years differ in issuer characteristics, but the distributions overlap substantially.
- Table IX tests whether banks in low fiscal-capacity countries are less able to issue safe CDs. The interaction terms are generally insignificant.
- Table X shows some local segmentation: French and German bank CD issuance responds more strongly to local domestic T-bill issuance than to foreign T-bill issuance.
- The local effect does not depend strongly on market stress.

Economic message. Country fiscal capacity is not the main boundary of private safety. The market treats short bank CDs as an asset class, with some domestic segmentation but little evidence that fiscal capacity explains safety.

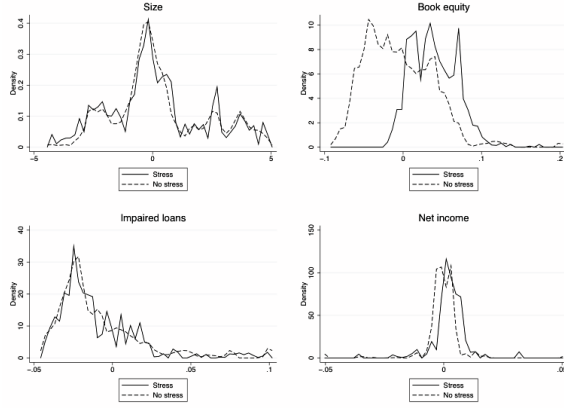


Figure 5: Cross-sectional characteristics of CD issuers, conditional on market stress. This figure plots the kernel distribution of four balance sheet characteristics (size, book equity, impaired loans, and net income) for CD issuers, conditional on market stress.

Figure 3: Issuer characteristics by stress state

Table IX
T-Bill Issuance and CD Issuance: Cross-Country Variation

This table presents results from estimation of the effect of T-bill issuance on CD issuance, after including cross-country variation in the effect. Regressions are at the issuer level. The measure of public asset issuance is the sum of T-bill issuances by France and Germany. The measure of private asset issuance includes all CD issuers in Europe. Regressions include issuer-date fixed effects and thus isolate variation across maturities for a given issuer. The baseline coefficients are interacted with a *LowCapacity* dummy that takes the value of one if the country of the issuing bank has an average CDS spread above 200 bps during the sample period. Standard errors are clustered at the issuer and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: $\Delta \log(1 + Q_{CD,i,mat,t})$		
$\Delta Q_{TB,i} \cdot \mathbb{1}(mat \leq 1w)$	-2.890** (1.314)	-4.010*** (1.374)	-4.085*** (1.395)
$\Delta Q_{TB,i} \cdot \mathbb{1}(mat \leq 1w) \cdot LowCapacity$	-0.857 (3.154)	0.940 (3.617)	0.669 (3.732)
$\Delta Q_{TB,i} \cdot \mathbb{1}(1w < mat \leq 1m)$	-1.544* (0.865)	-2.664*** (0.971)	-2.533** (0.998)
$\Delta Q_{TB,i} \cdot \mathbb{1}(1w < mat \leq 1m) \cdot LowCapacity$	2.584 (3.019)	4.382 (3.233)	4.080 (3.181)
$\Delta Q_{TB,i} \cdot \mathbb{1}(1m < mat \leq 3m)$	0.650 (0.692)	-0.469 (0.689)	-0.274 (0.697)
$\Delta Q_{TB,i} \cdot \mathbb{1}(1m < mat \leq 3m) \cdot LowCapacity$	1.781 (1.688)	3.579 (2.327)	3.500 (2.329)
R^2	0.000	0.252	0.252
Obs.	396,384	396,384	396,384
Issuer*Date FE	No	No	Yes
Issuer*Date FE	No	Yes	Yes

Table 6: Cross-country variation

Table X
T-Bill Issuance and CD Issuance: Response to Domestic T-Bill Supply

This table presents results from estimation of the effect of T-bill issuance on CD issuance, after including cross-country variation in the effect. Regressions are at the issuer level. The measure of public asset issuance is the sum of T-bill issuances by France and Germany. The measure of private asset issuance includes all CD issuers in Europe. Regressions include issuer-date fixed effects and thus isolate variation across maturities for a given issuer. We regress the change in short-term CDs, defined as CDs with an initial maturity below seven days, on T-bill issuance in four countries. This coefficient is interacted with a *Local* dummy equal to one if the bank is located in the T-bill issuing country. In some specifications, we further interact this coefficient with the VIX. Standard errors are clustered at the issuer and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: $\Delta \log(1 + \text{Short Term } Q_{CD,i,t})$							
	Country c:							
	France		Germany		Italy		Spain	
$\Delta Q_{TB,i,t}$	-2.188*** (0.604)	-4.334*** (1.142)	-0.134* (0.088)	-0.145* (0.090)	-2.791 (1.925)	-2.314 (1.879)	0.451 (0.527)	0.149 (0.207)
$\Delta Q_{TB,i,t} \cdot Local_c$	-1.866*** (0.449)	-2.672 (1.786)	-1.811*** (0.250)	-2.351*** (0.0275)	4.051*** (1.362)	3.561*** (1.317)	-1.925** (0.645)	-2.910* (1.633)
$\Delta Q_{TB,i,t} \cdot Local_c \cdot HighVIX_t$		1.972 (1.761)		0.569 (0.621)		0.670 (0.722)		-6.292 (5.123)
R^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Obs.	88,434	88,426	88,434	88,426	88,434	88,426	88,434	88,426
Issuer*Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 7: Response to domestic T-bill supply

5.6 Table XI, Table XII, and Figure 6: other assets and alternative identification

Read:

- Table XI shows that the substitution relation is strongest for bank CDs and weaker or absent for nonfinancial commercial paper and repos.
- In France, bank CDs fall with French T-bill issuance, with a coefficient of -0.364 for banks and -1.191 for local banks.
- In Germany, bank CDs also fall with German T-bill issuance, with a coefficient of -0.331 .
- Repos do not behave like private safe substitutes; in some specifications they increase with T-bill issuance.
- Table XII uses T-bill bid-to-cover ratios as an IV for CD safety premia. The first stage is negative and significant, and the second stage shows that instrumented CD safety premia predict more short-CD issuance.
- Figure 6 shows more CD issuance around T-bill auction days during high bid-to-cover weeks.

Economic message. Bank CDs are the main private safe-asset substitute in this setting. The IV and auction-day results reinforce the interpretation that excess demand for public safety induces private short-CD

supply.

Table XI
Issuance of T-Bills and Other Short-Term Securities

In this table, we regress the issuance of various short-term securities in either France or Germany on T-bill issuance in either France or Germany. All data are from the national financial accounts. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: Short-Term Negotiable Securities Issued By:						Repos
	All Issuers (incl. gov.)	All Private Issuers	Banks (CDs)	Local Banks Only (CDs)	Other Financials	Firms (CP)	
	Panel A: Direct Effects: French Private Assets on French T-Bills						
ΔQ_{TBI}	0.632** (0.267)	-0.367 (0.267)	-0.364* (0.217)	-1.191*** (0.233)	-0.099** (0.042)	0.064 (0.053)	0.693 (0.568)
Freq.	Quarter	Quarter	Quarter	Quarter	Quarter	Quarter	Quarter
R^2	0.984	0.967	0.961	0.953	0.979	0.873	0.981
Obs.	96	96	96	96	96	96	96
FE	Y	Y	Y	Y	Y	Y	Y
Panel B: Direct Effects: German Private Assets on German T-Bills							
ΔQ_{TBI}	1.082*** (0.212)	0.082 (0.212)	-0.331*** (0.104)		-0.023 (0.021)	-0.009 (0.061)	0.773*** (0.206)
Freq.	Quarter	Quarter	Month		Quarter	Quarter	Month
R^2	0.964	0.880	0.960		0.914	0.935	0.908
Obs.	84	84	346		84	84	250
FE	Y	Y	Y		Y	Y	Y
Panel C: Cross-Country Effects: French Private Assets on German T-Bills							
ΔQ_{TBI}	0.628 (0.455)	-0.207 (0.449)	0.003 (0.368)	-0.594 (0.448)	-0.033 (0.073)	0.041 (0.089)	2.690*** (0.899)
Freq.	Quarter	Quarter	Quarter	Quarter	Quarter	Quarter	Quarter
R^2	0.983	0.967	0.960	0.938	0.978	0.871	0.983
Obs.	96	96	96	96	96	96	96
FE	Y	Y	Y	Y	Y	Y	Y
Panel D: Cross-Country Effects: German Private Assets on French T-Bills							
ΔQ_{TBI}	0.447*** (0.141)	0.125 (0.127)	-0.073 (0.064)		0.008 (0.013)	-0.023 (0.036)	-0.030 (0.118)
Freq.	Quarter	Quarter	Month		Quarter	Quarter	Month
R^2	0.956	0.882	0.924		0.913	0.935	0.903
Obs.	84	84	286		84	84	250
FE	Y	Y	Y		Y	Y	Y

The respective substitution rates are 63% and 107%. Second, we find that in

Table 8: T-bill issuance and other short-term securities

Table XII
T-Bill Issuance and CD Issuance: IV Estimation

In this table, we regress changes in log short-term CDs outstanding on the CD safety premium, where the CD safety premium is instrumented using the T-bill bid-to-cover ratio (BTC). Panel A displays descriptive statistics on the BTC and its two components (bid = T-bill demand and cover = T-bill supply). In Panel B, we study the dynamics of the instrument by regressing it on monthly flows into European money market funds. Panel C shows the regression estimates for the two stages of the instrumental variables estimation. The CD safety premium is computed using equation (1), in which we use the Eonia swap rate as the risk-free rate, r_f . Observations are at a weekly frequency. Short-term CDs are defined as having a maturity less than or equal to seven days at issuance. Control variables include: $\log(Q_{CD,t-1})$, $\Delta \log(Q_{CD,t-1})$, total assets, return on assets, loans over assets, customer deposits over assets, common equity over assets, and impaired loans over total loans. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Descriptive Statistics											
	Min	10pc	25pc	Median	Mean	75pc	90pc	Max	Std.	Obs.	
Bid-to-cover ratio	1.62	2.14	2.37	2.67	2.69	2.94	3.36	4.41	0.47	358	
Bid (EUR billion)	6.20	14.80	17.60	20.60	20.40	23.20	26.60	42.30	4.70	358	
Cover (EUR billion)	2.90	6.40	7.30	8.00	7.90	8.60	9.60	11.50	1.40	358	
Fund flows (EUR billion)	-77.80	-24.30	-15.00	-4.00	-3.40	8.30	15.50	81.50	19.80	83	
Panel B: Variation in the Instrument											
$Flow_t$	ΔBTC_t 0.344* (0.161)				ΔBTC_t 0.708* (0.337)				$\Delta \log(Bids_t)$ 1.837* (0.852)		
$\log(Cover_t)$	-0.003 (0.193)				-0.096 (0.317)						
R^2	0.007				0.009				0.011		
Observations	83				83				83		
FE	-				YQ				YQ		
Panel C: Instrumental Variables Estimation											
	<i>First Stage</i>				<i>Second Stage</i>						
					Dependent Variable:						
	$SP_{CD,t}$				$\Delta \log(\text{Short term } Q_{CD,i,t})$						
Bid-to-cover ratio _t	-0.006*** (0.002)				-0.005*** (0.002)						
Instrumented $SP_{CD,t}$									-8.187** (4.061)		-9.619* (5.315)
Controls	No				Yes				No		Yes
F-statistic	11.48				55.65				-		-
R ²	0.010				0.011				-		-
Observations	14,725				14,649				14,725		14,649

Table 9: IV estimation using bid-to-cover ratios

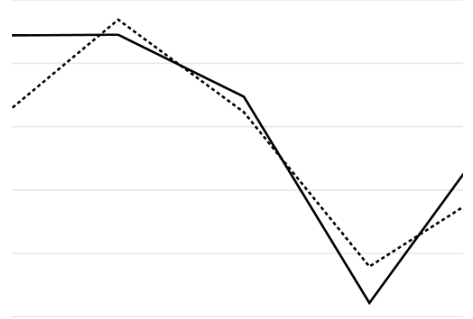


Figure 4: CD issuance on T-bill auction days

6 Short summary

- The paper studies whether private financial institutions can produce safe assets and when this private safety breaks down.
- The empirical laboratory is the European euro-denominated short-term debt market from 2008 to 2014.
- The key private asset is the certificate of deposit issued by banks.
- The main public benchmark is French and German T-bills.
- A safety premium is measured as the asset interest rate minus the Eonia swap rate of the same maturity.
- Only one-week CDs carry a safety premium, about -8.1 bps per year on average.
- Longer CDs carry positive spreads over the risk-free benchmark and are therefore not safe in the paper's exact sense.
- Banks increase the issuance of very short CDs when the supply of French and German T-bills falls.
- A 1% decrease in public safe asset supply is associated with roughly a 4% increase in very short CD issuance in the baseline fixed-effect specification.

- The same response is not found when Italian and Spanish T-bills are used as the public asset measure.
- During high-stress periods, the private-public substitution relation disappears.
- Aggregate CD market volumes do not collapse during stress; rather, the perceived substitutability of CDs with T-bills disappears.
- Issuer balance-sheet strength and domicile-country fiscal capacity do not strongly preserve private safety in stress.
- Other private short-term assets do not show the same clean safe-asset substitution pattern as bank CDs.
- IV evidence using T-bill bid-to-cover ratios and auction-day timing supports the demand-driven interpretation.
- The paper’s central conclusion is that private safety exists, but only within narrow maturity and market-condition boundaries.

7 Conclusion

7.1 Core conclusion

Kacperczyk, Pérignon, and Vuillemeys show that private institutions can produce safe assets, but the ability is highly constrained. In their European CD market setting, private safety is concentrated in very short bank debt and vanishes during stress.

7.2 Economic intuition

Very short CDs can be information-insensitive because their short maturity makes them effectively senior and relatively insensitive to new information. When public safe assets are scarce, investors seek close substitutes, and banks issue more short CDs. But when aggregate uncertainty rises, even short private bank debt becomes information-sensitive, so investors stop treating it as a substitute for T-bills.

7.3 Takeaway

The private production of safe assets is possible but fragile. It does not create a stable substitute for public safe assets across all states of the world. This has policy relevance because private safe-asset production may reduce scarcity in normal times, but it cannot be relied on to provide safety when stress makes information sensitivity salient.